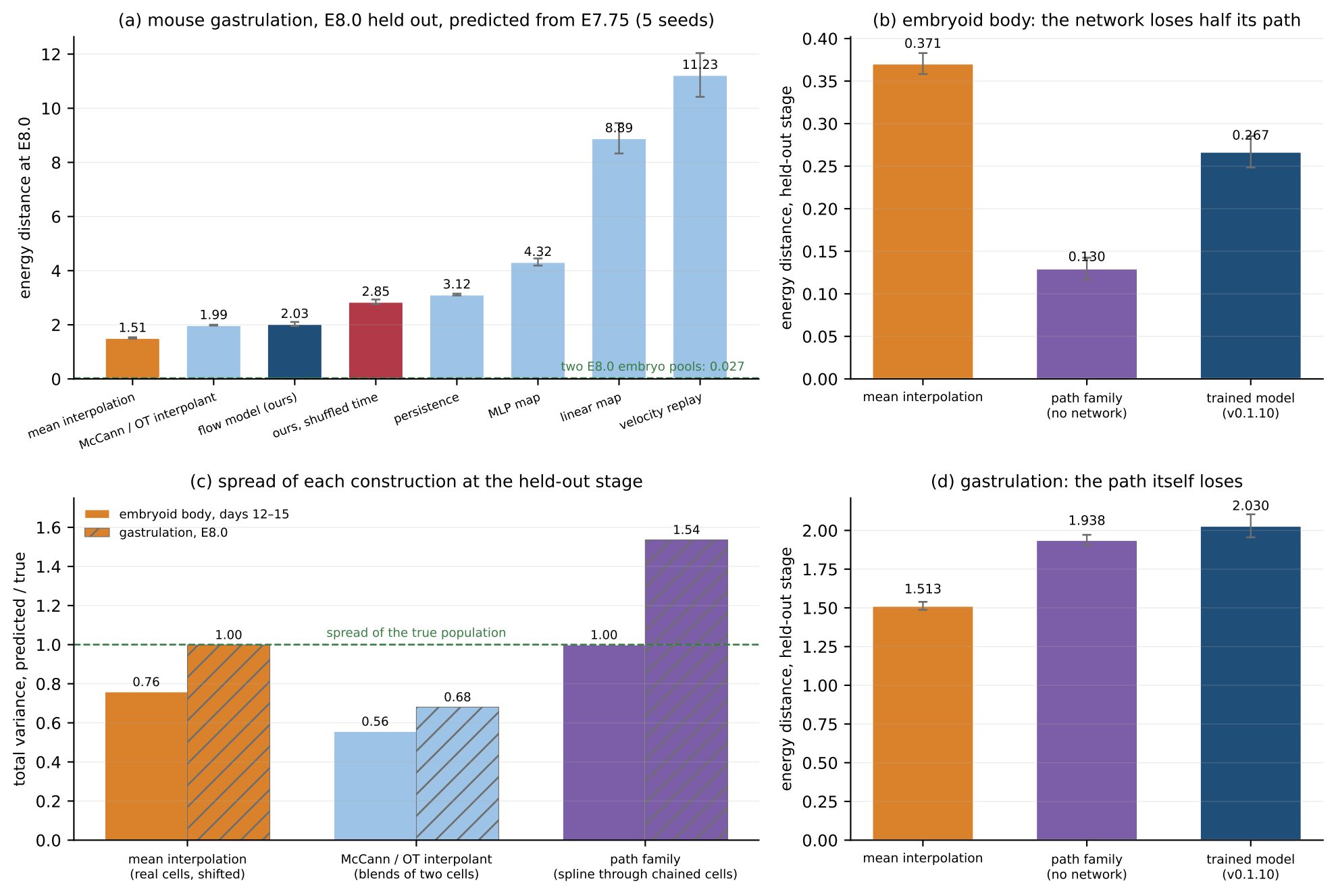
(a) mouse gastrulation, E8.0 held out, predicted from E7.75 (5 seeds)
energy distance at E8.0
1.51
1.99
2.03
2.85
3.12
4.32
8.89
11.23
two E8.0 embryo pools: 0.027
mean interpolation
McCann / OT interpolant
flow model (ours)
ours, shuffled time
persistence
MLP map
linear map
velocity replay
(b) embryoid body: the network loses half its path
energy distance, held-out stage
0.371
0.130
0.267
mean interpolation
path family (no network)
trained model (v0.1.10)
(c) spread of each construction at the held-out stage
embryoid body, days 12–15
gastrulation, E8.0
total variance, predicted / true
spread of the true population
0.76
1.00
0.56
0.68
1.00
1.54
mean interpolation (real cells, shifted)
McCann / OT interpolant (blends of two cells)
path family (spline through chained cells)
(d) gastrulation: the path itself loses
energy distance, held-out stage
1.513
1.938
2.030
mean interpolation
path family (no network)
trained model (v0.1.10)